

T.A.L.O.N.

Server User Guide

Installation, setup, Reticulum configuration, and server desktop operation.

Updated May 2, 2026 from the current PySide6 desktop code path.

Screenshots use isolated sample data and do not show live operational records.

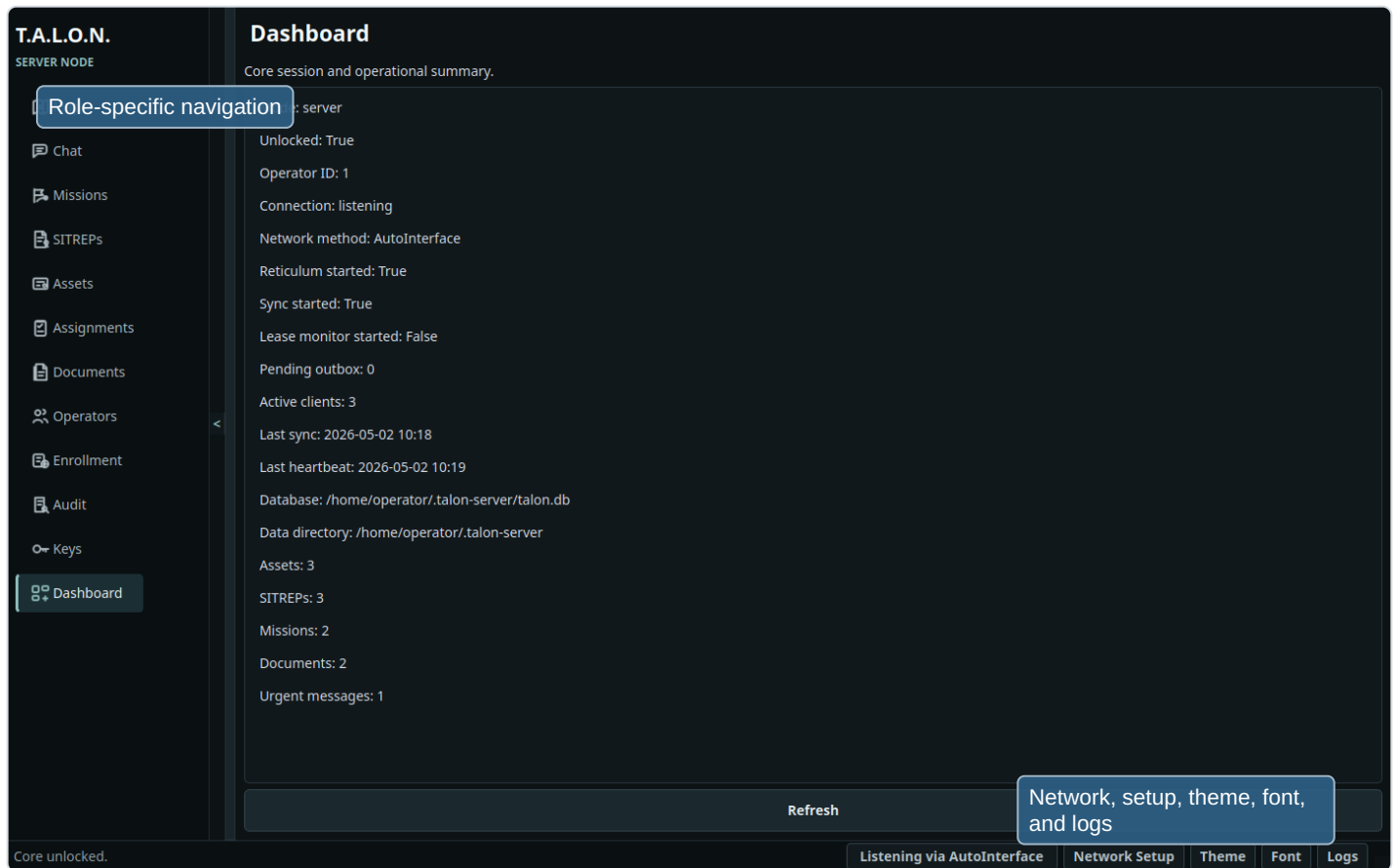
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3. First Run And Reticulum Setup
4. Main Shell
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Overview

T.A.L.O.N. is a tactical awareness and linked operations network for resilient coordination when normal infrastructure is unavailable. The server desktop is the command node for a deployment. It owns local server authority, client enrollment, operator leases, revocation, audit review, key status, and server-only operational actions.

The server uses a local SQLCipher database and syncs through Reticulum. Operators use the PySide6 desktop to manage maps, SITREPs, assets, missions, assignments, chat, documents, and operator records from one role-specific application.



Server desktop after unlock.

Installation

Install The Server Artifact

Use the server-specific Linux artifact. The installer reads the role marker from the extracted bundle and creates a server launcher, desktop menu entry, and desktop shortcut using the bundled TALON logo. Do not pass a role flag to the installer.

```
tar -xzf talon-desktop-server-linux.tar.gz
cd talon-desktop-server-linux
bash ./install.sh --yes
```

Installed Paths

Item	Default path or name
Launcher	<code>~/local/bin/talon-desktop-server</code>
Desktop menu entry	<code>\${XDG_DATA_HOME:-~/local/share}/applications/talon-desktop-server.desktop</code> , displayed as T.A.L.O.N. Server
Desktop shortcut	<code>~/Desktop/talon-desktop-server.desktop</code>

Data directory	~/.talon-server
Application config	~/.talon-server/talon.ini
TALON Reticulum config	~/.talon-server/reticulum/config
Document storage	~/.talon-server/documents
Launcher log	\${XDG_STATE_HOME:-~/local/state}/talon/desktop-server-launcher.log

Useful Installer Options

Option	Use
--yes	Assume yes for supported system package managers.
--no-deps	Skip system runtime dependency installation.
--no-bin	Skip role-specific launcher wrapper creation.
--no-desktop	Skip desktop menu entry and desktop shortcut creation.
--smoke-test	Run the installed package smoke test.
--uninstall --confirm-delete "DELETE TALON DATA"	Remove local TALON desktop files, data, identities, documents, logs, and launchers.

Role switch warning: a local user profile should not host both roles. Opposite-role cleanup and full uninstall require the exact phrase `DELETE TALON DATA`.

First Run And Reticulum Setup

Unlock Or Create The Local Database

Launch `talon-desktop-server`. On a fresh server with no local database, enter a new passphrase, enter it again to confirm it, review the on-screen passphrase requirements, and select **Create Passphrase**. Existing servers enter the server database passphrase and select **Unlock**.

T.A.L.O.N. SERVER Desktop

Create the local SQLCipher database passphrase.

New passphrase

Confirm passphrase

Requirements

- At least 12 characters.
- Include at least three of: lowercase letters, uppercase letters, numbers, symbols.
- Avoid common words, keyboard or number sequences, repeated characters, and predictable patterns.

Create Passphrase

Network setup is available after unlock.

Server first-run passphrase setup window.

Configure Reticulum

Network setup is available only after the encrypted database is unlocked. The server uses a TALON-owned Reticulum config under the server data directory. Use the dialog to pick a template, import a config, edit the raw Reticulum stanza, validate, then save or continue.

Reticulum Configuration
TALON Reticulum config: /home/operator/.talon-server/reticulum/config

TALON Default
AutoInterface
TCP Server
TCP Client

Yggdrasil Server
Yggdrasil Client
i2pd Server
i2pd Client

Import ~/.reticulum/config

TALON-owned config path

Templates for common transports

```

[reticulum]
enable_transport = true
share_instance = No

[interfaces]
[[TALON AutoInterface]]
type = AutoInterface
enabled = Yes

# Add TCP, Yggdrasil, i2pd, or RNode stanzas for deployment.

```

Reticulum config is valid.

Validate before continuing

Validate
Cancel
Save
Continue

Reticulum setup after SQLCipher unlock.

- **TALON Default** creates a role-appropriate baseline with `share_instance = No`.
- **AutoInterface** is useful for local mesh discovery where supported.
- **TCP Server** listens for TCP peers. Use direct TCP only when its metadata exposure is acceptable or protected by another layer.
- **Yggdrasil Server** and **i2pd Server** generate stanzas for those overlay paths.
- If Reticulum is already running, saved changes require restarting TALON before they affect sync.

Main Shell

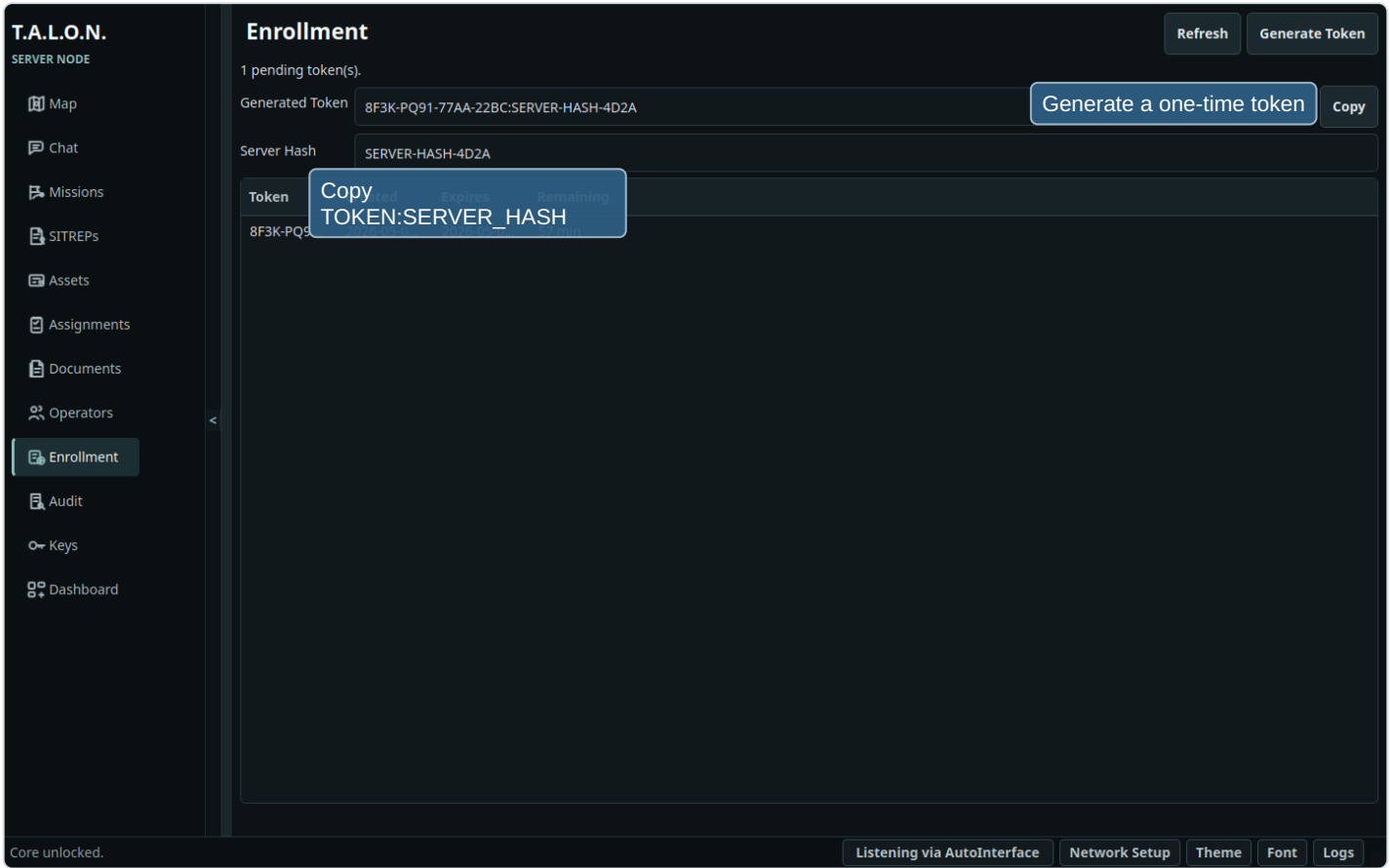
The server main window has a collapsible navigation rail, the active page, and a status bar. The server navigation contains Map, Chat, Missions, SITREPs, Assets, Assignments, Documents, Operators, Enrollment, Audit, Keys, and Dashboard. The status bar shows core unlock state, the redacted network method, Network Setup, Theme, Font, and Logs.

Unread badges: red-dot indicators appear for relevant updates from other operators. The Map and Dashboard sections do not show red dots.

Server Administration

Enrollment

Use Enrollment to generate one-time client enrollment tokens. Give the full `TOKEN:SERVER_HASH` value to the client operator through a trusted out-of-band channel. Tokens are temporary and disappear after use or expiry.



Enrollment token generation and pending-token review.

Operators

The Operators page lists enrolled operators, lease state, profile details, skills, and revocation state. Server operators can edit profiles, renew leases, and revoke access after confirmation.

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SERVER NODE

Map

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Missions

SITREPs

Assets

Assignments

Documents

Operators

Enrollment

Audit

Keys

Dashboard

Operators

3 operator(s), 3 active, 0 locked, 0 revoked.

ID	Callsign	Status	Role	Skills	Lease
10	ECHO-2	OK (48h)	Field Oper...	medical, n...	2026-05-04 11:00
11	BRAVO-4	OK (48h)	Team Lead	comms, lo...	2026-05-04 11:00
12	DELTA-1	OK (48h)	Support	mechanic, ...	2026-05-04 11:00

#10 ECHO-2

Display name: ECHO 2

Role: Field Operator

Status: OK (48h)

Enrolled: 2026-05-01 11:00

Lease expires: 2026-05-04 11:00

Skills: medical, navigation

RNS hash: echo-2-rns-hash-0123456789abcdef

Sample operator profile for guide screenshots.

Core unlocked.

Listening via AutoInterface

Network Setup

Theme

Font

Logs

Operator roster and lease controls.

Display Name

ECHO 2

Role

Field Operator

Notes

Sample operator profile for guide screenshots.

Skills

☐ medic

☐ comms

☐ logistics

☐ intelligence

☐ recon

☒ navigation

☐ engineering

☐ security

Custom Skills

medical

Custom skill

Add

Remove

Cancel

Save

Profile and skills editor.

Audit And Keys

Audit shows recent server audit events and payload detail. Keys shows the server hash, Reticulum identity state, operator key status, and server revocation controls. The visible group-key rotation control reports current implementation status and does not silently rotate keys.

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Audit

Event filterSearchClear

3 audit entrie(s).

ID	Occurred	Event	Payload
1001	2026-05-0...	enrollmen...	operator=COMMAND
1002	2026-05-0...	missions.a...	assets=201,203 mission_id=301
1003	2026-05-0...	operators....	operator=ECHO-2

#1001 enrollment.token.created
Occurred: 2026-05-02 10:00
operator: COMMAND

Core unlocked.

Listening via AutoInterfaceNetwork SetupThemeFontLogs

Audit event search and detail view.

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Keys

RefreshRotate Group KeyRevoke

3 operator identity record(s), 0 revoked.

Server Hash

SERVER-HASH-4D2A

Identity Status

Reticulum started

ID	Callsign	Status
10	ECHO-2	OK (48h)
11	BRAVO-4	OK (48h)
12	DELTA-1	OK (48h)

Core unlocked.

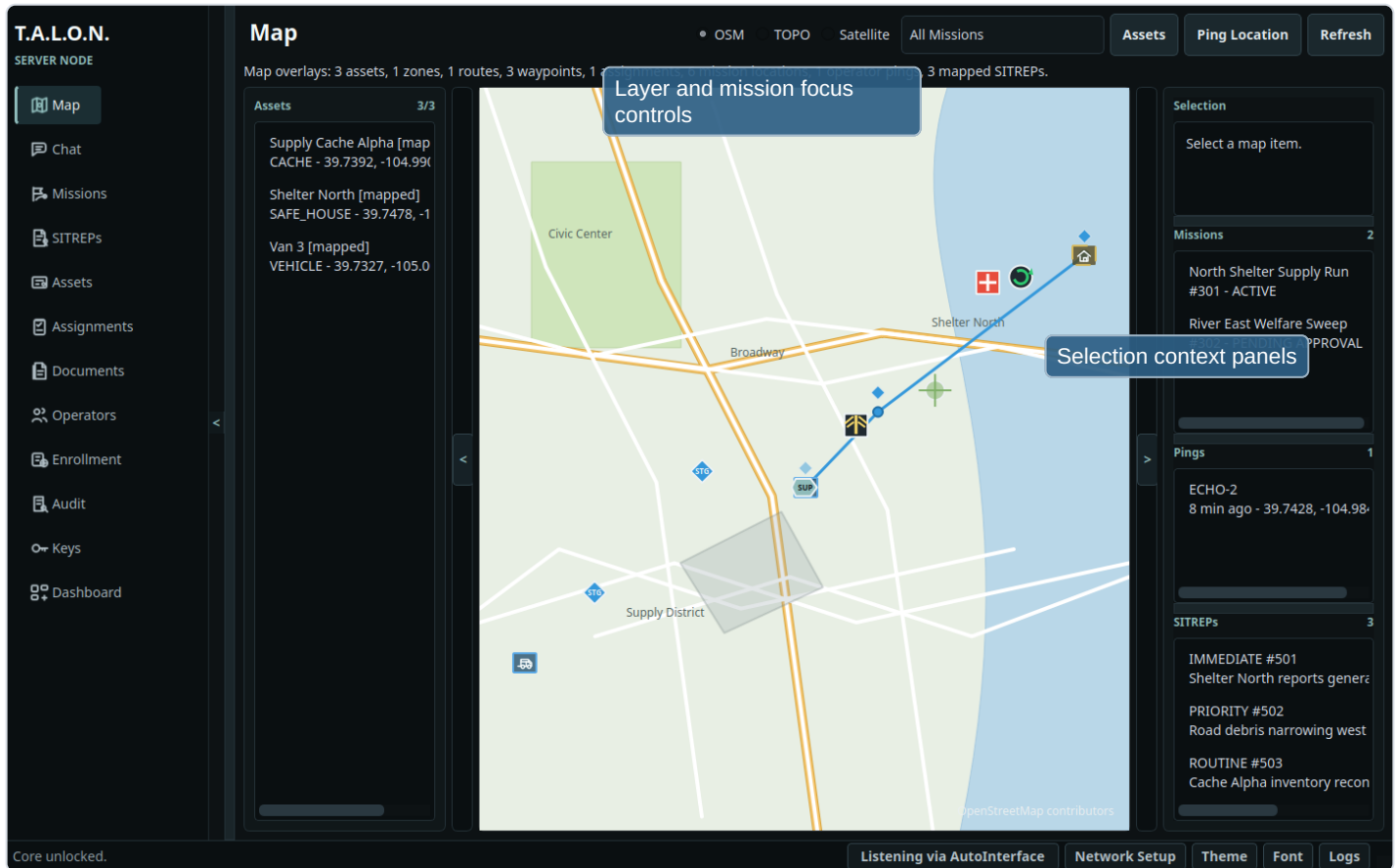
Listening via AutoInterfaceNetwork SetupThemeFontLogs

Server hash, identity status, and operator key status.

Screen Walkthrough

Map

The Map page displays OSM, TOPO, or Satellite base-layer controls with TALON overlays for assets, mission areas, routes, waypoints, assignments, operator pings, and located SITREPs. If an operator will be in an area with no connectivity, or limited connectivity, map tiles can be pre cached.



Operational map with fallback OSM-style background.

SITREPs

SITREPs provide a severity feed, open/closed filtering, append-only follow-up notes, assignment, server-only delete, and opt-in FLASH/FLASH_OVERRIDE audio. New SITREPs can include mission, assignment, asset, and map-picked location context.

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SITREPs

New SITREP

Audio

Refresh

ROUTINE

0

PRIORITY

1

IMMEDIATE

1

FLASH

0

FLASH OVERRIDE

0

Pending

0

Level

All levels

Show closed

SITREP List

IMMEDIATE / BRAVO-4

2026-05-02 10:30

Shelter North reports generator fuel below two hours.

Location: Shelter North | Asset: Shelter North | Mission

Open

PRIORITY / ECHO-2

2026-05-02 10:45

Road debris narrowing west approach. Use east route.

Location: Broadway Checkpoint | Assignment | Mission

Open

Selected Detail

Open

IMMEDIATE SITREP #501 / DELTA-1

BRAVO-4 | 2026-05-02 10:30 | Open | Shelter North | Assigned: DELTA-1 | Mission: North Shelter Supply Run | Asset: Shelter North

Shelter North reports generator fuel below two hours.

Add information

Assign

Append

Server-only destructive control

Delete

2026-05-02 10:40 Assigned

Assigned to DELTA-1. DELTA-1 checking fuel

2026-05-02 10:50 Note

Shelter has one spare can but needs delivery

Core unlocked.

Listening via AutoInterface

Network Setup

Theme

Font

Logs

SITREP feed and selected report detail.

Level

ROUTINE

Body

Shelter North reports generator fuel below two hours.

✓ Optional context

Mission

None

Assignment

None

Asset

None

Location

Shelter North

Lat / Lon

Latitude

Longitude

Pick on Map

Use Asset Position

Clear

• OSM

• TOPO

• Satellite

Click the map to set point.

Clear

Undo

Use Selected

Send

Cancel

New SITREP dialog with optional context and embedded map picker.

Assets

Assets can be created and edited with category, label, description, and coordinates. Server mode can verify or unverify assets and can hard-delete after confirmation. Client deletion requests appear in the asset list.

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Dashboard

Assets

AllRefreshNewEditUnverifyDelete

3 assets, 2 verified, 1 deletion request(s).

ID	Label	Category	Verified	Delete	Mission	Location
201	Supply Ca...	Cache	Yes		301	39.739200,...
202	Shelter No...	Safe House	No	Requested		39.747800,...
203	Van 3	Vehicle	Yes		301	39.732700,...

#201 Supply Cache Alpha

Category: Cache

Verified: Yes

Created by: 10

Confirmed by: 1

Mission: 301

Location: 39.739200, -104.990300

Deletion requested: No

Water, medical, and radio batteries.

Core unlocked.

Listening via AutoInterfaceNetwork SetupThemeFontLogs

Asset table, verification state, and detail panel.

Category	Cache
Label	Supply Cache Alpha
Description	Water, medical, and radio batteries.
Latitude	39.7392
Longitude	-104.9903
Map	Pick on Map

Cancel

Save

Asset create/edit dialog with typed or map-picked coordinates.

Missions

Missions include title, type, priority, team lead, team members, requested assets, AO polygon, route, timing, support notes, constraints, objectives, and key locations. Server lifecycle controls include approve, reject, abort, complete, delete, and server edit. Approval opens asset allocation review.

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Missions

AllRefreshNew MissionEdit MissionApproveRejectAbortCompleteDelete

2 mission(s).

ID	Title	Status	Priority	Type
301	North Shel...	Active	PRIORITY	Supply Run
302	River East ...	Pending A...	IMMEDIATE	Welfare Check

#301 North Shelter Supply Run

Status: active

Priority: PRIORITY

Type: Supply Run

Creator: COMMAND

Channel: mission-301

Team Lead: BRAVO-4

Team Members: ECHO-2

Activation: 2026-05-02 14:00

Window: 2026-05-02 14:00 - 18:00

Max Duration: 4 hours

Staging Area: 39.739800, -104.995000

Demob Point: 39.747000, -104.980500

Stand-down: Shelter confirms receipt.

Move supplies from Cache Alpha to Shelter North before night shift.

Constraints:

Avoid flooded underpass

Maintain radio check every 20 min

Phases:

Stage | Load supplies | 30 min

Deliver | Move to shelter | 90 min

Support:

Medical: Basic trauma kit staged at shelter.

Logistics: Two hand carts.

Comms: Primary: All Hands, backup: direct call.

Equipment: Vehicle straps and lamps.

Fuel | Van 3 ready

Objectives:

Delivery | All supplies checked in

Key Locations:

Command Post: 39.741500, -104.988000

Medical: 39.746800, -104.982000

Supply: 39.739200, -104.990300

Assets:

#301 Supply Cache Alpha

Core unlocked.

Listening via AutoInterfaceNetwork SetupThemeFontLogs

Mission list, status filter, detail, and lifecycle controls.

Create Mission

Cancel

Create

Overview

Timing

Assets

Area / Route

Support

AO Polygon

39.7364, -104.9960

39.7480, -104.9900

39.7460, -104.9770

Draw on Map

Route / Waypoints

39.7392, -104.9903

39.7420, -104.9870

39.7478, -104.9776

Draw on Map

Objectives

Label	Criteria

Add

Remove

Mission Location Icons

Command Post

Staging Area

Demob Point

Map target: AO Polygon.

OSM

TOPO

Satellite

3 point(s). Latest:
39.746000,
-104.977000

Clear

Undo

Use Selected

Mission create/edit workflow with embedded map drawing.

#302 River East Welfare Sweep

Requested by: ECHO-2

Priority: IMMEDIATE | Type: Welfare Check

Asset Allocation

☐ #202 Shelter North [safe_house]

Cancel

Approve

Asset allocation review before approval.

Assignments

The Assignment Board tracks assigned mission and SITREP follow-up work, operator availability, unassigned targets, and pending sync. The server can assign an available operator to an unassigned mission or SITREP and optionally message that operator.

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Assignment Board

All assigned work

Refresh

Assign Operator

2 assigned work item(s). Assigned mission and SITREP work is shown here; mission planning, SITREP resolution, and map placement remain on their dedicated pages.

Assigned Missions

1

active or pending approval

Assigned SITREPs

1

unresolved follow-ups

Unassigned Work

2

needs an operator

Available Operators

1

not committed elsewhere

Pending Sync

0

local assignment changes

Assigned Work

missions and SITREP follow-ups

Target

Shelter North reports generator fuel below two h

North Shelter Supply Run

Operator Status

perators are excluded from assignment picker

Operator	State	Current Work
BRAVO-4	Assigned	Deliver Cache Alpha
DELTA-1	Assigned	SITREP #501: Shelter North report...
ECHO-2	Assigned	Deliver Cache Alpha
COMMAND	Available	Server Authority

Selected Work

target details

Shelter North reports generator fuel below two hours.

Type: SITREP

SITREP: #501

Status: Open

Priority: IMMEDIATE

Assigned: DELTA-1

Mission: North Shelter Supply Run

Location: Shelter North

Body:

Shelter North reports generator fuel below two hours.

Recent follow-ups: 2

10:40 - assigned

10:50 - note

Open SITREP

Message Operator

Reassign

Unassigned Work

next assignment candidates

River East Welfare Sweep

IMMEDIATE - Mission - Welfare Check | created 1 h

Road debris narrowing west approach. Use east roi

PRIORITY - SITREP - PRIORITY | created 15 min ago

Core unlocked.

Listening via AutoInterface

Network Setup

Theme

Font

Logs

Assigned work, operator status, selected detail, and unassigned work.

Unassigned Targets

Sort

Priority

Target

River East Welfare Sweep

Road debris narrowing west approach. Use east route

Unassigned Operators

Operator

Status

Skills

Selection Summary

Select one unassigned target and one available operator.

Cancel

Assign and Message

Assign an available operator to unassigned mission or SITREP work.

Chat

Chat includes grouped channels, default channels, custom channels, mission channels, direct-message channels, urgent messages, grid-reference attachments, operator presence, alert summaries, and server-only delete controls. Direct-message channels are server-readable until end-to-end encryption is implemented.

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Chat

4 channel(s), 2 message(s) here, 1 urgent.

Channels

Search channels

MISSION

mission-301

SQUAD

logistics

DIRECT

DM ECHO-2 / BRAVO-4

DEFAULT

all-hands

all-hands | Default

10:00 COMMAND

Operational period begins at 1400.

10:30 BRAVO-4 URGENT

Shelter North requesting fuel update.

Grid: 39.747800, -104.977600

Message

Compose message

☐ Urgent

Delete Message

Send

Operators

BRAVO-4 online Team Lead

ECHO-2 online Field Operator

DELTA-1 offline Support

Alerts

URGENT: Shelter North fuel up:

COMMAND | Server Authority

Core unlocked.

Listening via AutoInterface

Network Setup

Theme

Font

Logs

Channel list, message timeline, composer, operators, and alerts.

Operator

ECHO-2 | Field Operator | online

BRAVO-4 | Team Lead | online

DELTA-1 | Support | offline

Cancel

Open

Direct-message peer picker.

Search known positions

Type	Name	Latitude	Longitude	Detail
ASSET	Shelter No...	39.747800	-104.977600	Safe House #202
ASSET	Supply Ca...	39.739200	-104.990300	Cache #201
ASSET	Van 3	39.732700	-105.003100	Vehicle #203
SITREP	IMMEDIAT...	39.747800	-104.977600	Shelter North
SITREP	PRIORITY ...	39.742000	-104.987000	Broadway Checkpoint
SITREP	ROUTINE ...	39.739200	-104.990300	Linked to Supply Cache Alpha
WAYPOINT	Broadway ...	39.742000	-104.987000	Mission 301 waypoint 2
WAYPOINT	Cache Alpha	39.739200	-104.990300	Mission 301 waypoint 1
WAYPOINT	Shelter No...	39.747800	-104.977600	Mission 301 waypoint 3
ZONE	Flooded U...	39.736000	-104.992725	Hazard centroid #701

ASSET Shelter North 39.747800, -104.977600

10 known position(s).

Clear

Cancel

Attach

Attach a map position to a chat message.

Documents

Server mode can upload, organize, download, move, rename, and delete documents. Document validation is enforced by core handling. Macro-capable office formats are called out as higher risk before save.

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Refresh

Upload

2 document(s), 240.2 KB total, 1 macro-risk type(s). Server mode: context menus manage repository structure.

REPOSITORY		ALL DOCUMENTS				DETAILS
		NAME	SIZE	UPLOADER	UPLOADED	
All Documents	2	shelter-supply-list.pdf	177.7 KB	COMMAND	2026-05-02	shelter-supply-list.pdf Right-click this document for file actions. Drag it onto a folder to move it there. Foldermissions/shelter-north Size177.7 KB UploaderCOMMAND Uploaded2026-05-02 09:53 SHA-256d2b21ab92219efaa33cb2298... Current delivery manifest for Shelter North.
Root	0	generator-checklist.docm	62.5 KB	BRAVO-4	2026-05-02	
missions	0					
shelter-north	1					
operations	1					

Context actions are server-only. Clients keep download and detail actions only.

Rename accepts Enter, Escape cancels.

Core unlocked.

Listening via AutoInterfaceNetwork SetupThemeFontLogs

Repository folders, document list, and detail panel.

File

/home/operator/briefing.pdf

Browse

Folder

missions/shelter-north

Description

Mission packet and shelter manifest.

Cancel

Upload

Server upload dialog with target folder and description.

Security And Data Handling

Local Database

The server unlocks a local SQLCipher database. The passphrase derives the local key used for database and protected local material.

Network Identity

Reticulum identity material belongs in the TALON role data directory and is protected by the unlocked local state. Do not share server and client Reticulum directories.

Transport

TALON sync traffic uses Reticulum. The desktop reports only the transport family, not addresses, ports, peers, or device names.

Documents

Uploads are size-checked, filename-checked, extension-checked, MIME-checked where available, and hash-verified on download.

Direct TCP: direct TCP can reveal network addressing metadata. Use Yggdrasil, i2pd, LoRa, or a trusted VPN when address privacy matters.

Troubleshooting

Issue	Action
Application will not launch.	Run <code>talon-desktop-server --smoke --mode server</code> or reinstall with <code>--smoke-test</code> . Check the launcher log path listed above.
Network setup keeps appearing.	Unlock, open Network Setup, validate the TALON Reticulum config, save or continue, then restart if Reticulum was already running.
A client operator cannot enroll.	Confirm the server is running, the server Reticulum path is reachable, the client has the matching client transport config, and the full <code>TOKEN: SERVER_HASH</code> was entered before expiry.
Direct TCP warning appears.	The active Reticulum method is direct TCP. Change to Yggdrasil, i2pd, LoRa, or a trusted VPN-backed path if address privacy matters.
Document upload is blocked.	Check file size, extension, filename, MIME type, and whether the file is an executable or script type.